

What drives the possibility for PFAS to permeate cells: a MD study

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ABSTRACT

Per- and polyfluoroalkyl substances (PFAS) constitute a vast family of several thousand chemical compounds. Known for their non-stick, waterproof, and heat-resistant properties, these substances are widely used in textiles, packaging, fire-fighting foams, refrigerants, non-stick coatings, cosmetics, and more.

PFAS are associated with various adverse health effects, and it is crucial to better understand how these compounds interact with the human body. In particular, it is important to determine which physicochemical and structural parameters—such as size, charge, and hydrophobicity—influence the ability of PFAS to cross cellular membranes.

To address this, we employ molecular dynamics simulations aimed at identifying the factors that affect their passive permeation. The modeling protocol used (Memcross¹), based on the Accelerated Weight Histogram (AWH) algorithm and previously developed in our laboratory, allows us to track, at the atomic scale, the movements and insertion of the studied molecule into lipid bilayers.

In this preliminary study, we selected six perfluorinated acids in carboxylic or sulfonic form, which are frequently detected in drinking water, the primary source of human contamination.

REFERENCES

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